

Quantum Fourier transform

DFT:

$$F_N = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & & & 1 \\ \omega_N & & & \omega_N^{N-1} \\ \omega_N^2 & & & \omega_N^{2(N-1)} \\ \vdots & & & \vdots \\ 1 & \omega_N^{N-1} & & \omega_N^{(N-1)(N-1)} \end{bmatrix}$$

$$(F_N)_{kj} = \omega_N^{(k-1)(j-1)} \quad \begin{matrix} k=1 \dots N \\ j=1 \dots N \end{matrix}$$

$$\omega_N = e^{\frac{2\pi i}{N}} = \cos \frac{2\pi}{N} + i \sin \frac{2\pi}{N}$$

$$y = F_N x \quad y = \text{DFT}(x)$$

FFT are algorithms for computing DFT with $O(N \log N)$ flops

$$F_N \text{ is unitary} \quad F_N F_N^\dagger = I \quad \neg$$

$$\sum_{k=0}^{N-1} \omega_N^{jk} = \begin{cases} N & \text{if } j=0 \pmod{N} \\ 0 & \text{otherwise} \end{cases}$$

$$y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j e^{\frac{2\pi i j k}{N}}$$

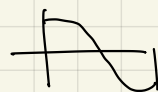
Rmk

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \text{ is } F_2$$

$$N=2$$

$$\omega_2 = \cos \frac{2\pi}{2} + i \sin \frac{2\pi}{2} = -1 + i \cdot 0$$

$$y = H^{\otimes n} |x\rangle \text{ is a QFT over } \mathbb{Z}_2 \otimes \mathbb{Z}_2 \otimes \dots \otimes \mathbb{Z}_2$$



Coppersmith in 1994 $\rightarrow O(\log^2 N)$ $N = 2^n$

$$|x\rangle = \sum_{j=0}^{N-1} x_j |j\rangle = x_0 |0\rangle + x_1 |1\rangle + \dots + x_{N-1} |N-1\rangle$$

$$\text{QFT}(|x\rangle) = \sum_{j=0}^{N-1} x_j \cdot \underset{\uparrow}{\text{QFT}}(|j\rangle)$$

$$|\tilde{j}\rangle = \text{QFT}(|j\rangle)$$

$$|\tilde{j}\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{2\pi i j k}{N}} |k\rangle$$

Ex: $n=1$ $N=2$ 1 qubit systems $\{|0\rangle, |1\rangle\} \rightarrow \{|+\rangle, |-\rangle\}$

$$|+\rangle = |\tilde{0}\rangle = \frac{1}{\sqrt{2}} \left(e^{\frac{2\pi i j 0}{2}} |0\rangle + e^{\frac{2\pi i j 1}{2}} |1\rangle \right) = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$|-\rangle = |\tilde{1}\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$$|\tilde{j}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{\pi i j} |1\rangle)$$

n qubit transform:

$$|\tilde{j}\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{2\pi i j k}{N}} |k\rangle$$

1. $|k\rangle = |k_1 k_2 \dots k_n\rangle$

$$\sum_{k=0}^{N-1} \rightarrow \sum_{k_1=0}^1 \sum_{k_2=0}^1 \dots \sum_{k_n=0}^1$$

$$k = k_1 k_2 k_3 \rightarrow k = \sum_{s=1}^3 k_s 2^{3-s}$$

$$\underbrace{101}_{k=1 \cdot 2^{3-1} + 0 \cdot 2^{3-2} + 1 \cdot 2^{3-3}}$$

$$|\tilde{j}\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{2\pi i j}{N} \left(\sum_{s=1}^n k_s 2^{n-s} \right)} |k_1 k_2 \dots k_n\rangle$$

$$= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i j \sum_{s=1}^n \frac{k_s}{2^s}} |k_1 k_2 \dots k_n\rangle$$

$$= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \left(\prod_{s=1}^n e^{\frac{2\pi i j k_s}{2^s}} \right) |k_1 \dots k_n\rangle$$

$$= \frac{1}{\sqrt{N}} \left(|0\rangle + e^{\frac{2\pi i j}{2}} |1\rangle \right) \otimes \left(|0\rangle + e^{\frac{2\pi i j}{2^2}} |1\rangle \right) \otimes \dots \otimes \left(|0\rangle + e^{\frac{2\pi i j}{2^n}} |1\rangle \right)$$

$$|j\rangle = |\bar{j}_1 \dots \bar{j}_n\rangle = |j_1\rangle \otimes |j_2\rangle \otimes \dots \otimes |j_n\rangle$$

$$|\tilde{j}\rangle = \frac{1}{\sqrt{N}} \left(|0\rangle + e^{\frac{2\pi i j}{2}} |1\rangle \right) \otimes \left(|0\rangle + e^{\frac{2\pi i j}{2^2}} |1\rangle \right) \otimes \dots \otimes \left(|0\rangle + e^{\frac{2\pi i j}{2^n}} |1\rangle \right)$$

$$|j_s\rangle \rightarrow \left(|0\rangle + e^{\frac{2\pi i j}{2^s}} |1\rangle \right)$$

$$|0 \dots 0\rangle \rightarrow \frac{1}{\sqrt{2}}$$

$$|00 \dots 1\rangle \rightarrow \frac{1}{\sqrt{2}} e^{\frac{2\pi i j}{2^n}}$$

$$|10 \dots 0\rangle \rightarrow \frac{1}{\sqrt{2}} e^{\frac{2\pi i j}{2}}$$

$$|0 \dots 011\rangle \rightarrow \frac{1}{\sqrt{2}} \left(e^{\frac{2\pi i j}{2^{n-1}} + \frac{2\pi i j}{2^n}} \right) = \frac{1}{\sqrt{2}} e^{2\pi i \left(\frac{j}{2^{n-1}} + \frac{j}{2^n} \right)}$$

$$j = \sum_{k=1}^n 2^{n-k} j_k$$

$$\frac{j}{2^s} = \sum_{k=1}^n 2^{n-k-s} j_k$$

$$\frac{j_1 \dots j_n}{2^s} = (j_1 \dots j_{n-s} \cdot j_{n-s+1} \dots j_n)$$

$$e^{\frac{2\pi i j}{2^s}} = \underbrace{e^{2\pi i (\underbrace{j_1 \dots j_{n-s}}_{\in \mathbb{N}})}}_1 e^{2\pi i (\underbrace{0 \cdot j_{n-s+1} \dots j_n}_{\in \mathbb{Q}})}$$

$$\begin{aligned} & \cos(2\pi (\underbrace{j_1 \dots j_{n-s}}_1)) + i \sin(2\pi (\underbrace{j_1 \dots j_{n-s}}_1)) \\ &= e^{2\pi i (\underbrace{0 \cdot j_{n-s+1} \dots j_n}_0)} \end{aligned}$$

$$|j_s\rangle \rightarrow |0\rangle + e^{\frac{2\pi i (0 \cdot j_{n-s+1} \dots j_n)}{2^s}} |1\rangle$$

$$U_S = \begin{bmatrix} 1 & 0 \\ 0 & e^{\frac{2\pi i}{2^s}} \end{bmatrix}$$

$$U_S |0\rangle = |0\rangle$$

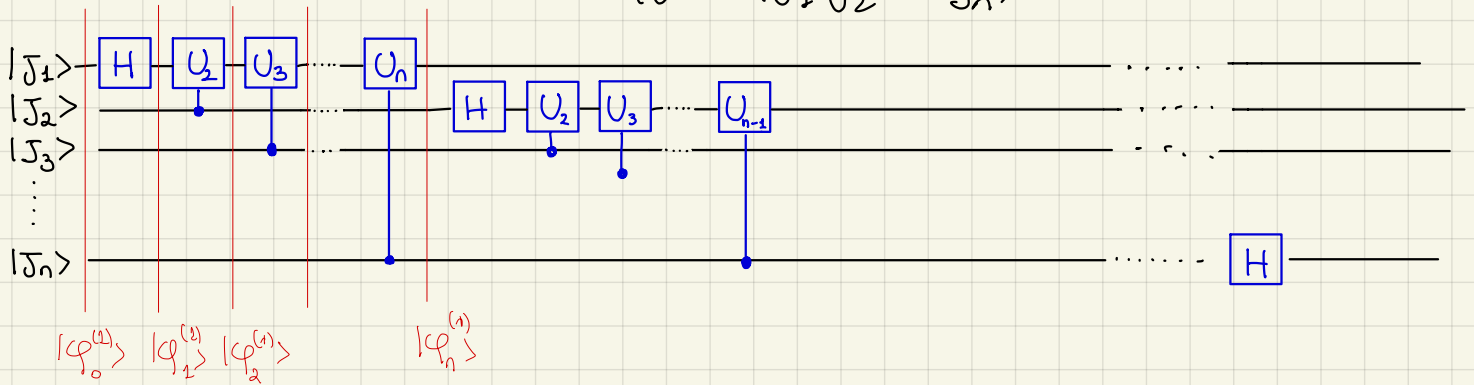
$$U_S |1\rangle = e^{\frac{2\pi i}{2^s}} |1\rangle$$

$$U_S \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$U_S \begin{bmatrix} p \\ 1 \end{bmatrix} =$$

The circuit

$$|j\rangle = |j_1 j_2 \dots j_n\rangle$$



$$|\varphi_0^{(n)}\rangle = |j_1 j_2 \dots j_n\rangle$$

$$|\varphi_1^{(n)}\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{\frac{2\pi i j_1}{2}} |1\rangle \right) |j_2 \dots j_n\rangle$$

$$U_2 = \begin{bmatrix} 1 & 0 \\ 0 & e^{\frac{2\pi i j_2}{2^2}} \end{bmatrix}$$

$$|\varphi_2^{(n)}\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{\frac{2\pi i j_1}{2}} e^{\frac{2\pi i j_2}{2^2}} |1\rangle \right) |j_2 \dots j_n\rangle$$

$$= \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i \left(\frac{j_1}{2} + \frac{j_2}{2^2} \right)} |1\rangle \right) |j_2 \dots j_n\rangle$$

$$|\varphi_3^{(n)}\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i \left(\frac{j_1}{2} + \frac{j_2}{2^2} + \frac{j_3}{2^3} \right)} |1\rangle \right) |j_2 \dots j_n\rangle$$

...

$$|\varphi_n^{(n)}\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i \left(\frac{j_1}{2} + \frac{j_2}{2^2} + \frac{j_3}{2^3} + \dots + \frac{j_n}{2^n} \right)} |1\rangle \right) |j_2 \dots j_n\rangle$$

$$e^{2\pi i \left(\frac{j_1}{2} + \frac{j_2}{2^2} + \dots + \frac{j_n}{2^n} \right)} = e^{\frac{2\pi i j}{2^n}}$$

$$|\varphi_n^{(n)}\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{\frac{2\pi i j}{2^n}} |1\rangle \right) |j_2 \dots j_n\rangle$$

For qubit 2

$$|\varphi_n^{(n)}\rangle = \frac{1}{(\sqrt{2})^n} \left(|0\rangle + e^{\frac{2\pi i j}{2^n}} |1\rangle \right) \left(|0\rangle + e^{\frac{2\pi i j}{2^{n-1}}} |1\rangle \right) \dots$$

$$\uparrow$$

$$\otimes \left(|0\rangle + e^{\frac{2\pi i j}{2}} |1\rangle \right)$$

I have to swap the order of qubits to get our QFT(j)

$$\text{Cost: } n + n-1 + \dots + 1 = \frac{n(n+1)}{2}$$

+ $\frac{n}{2}$ swaps

$O(n^2)$ gates vs $n \cdot 2^n$ flops for classical

$$\text{QFT}(|x\rangle) = \sum_{j=0}^{N-1} x_j \cdot \text{QFT}(|j\rangle) =$$

$$= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j \left(\sum_{k=0}^{N-1} e^{\frac{2\pi i j k}{N}} |k\rangle \right)$$

$$\uparrow$$

$$\frac{1}{\sqrt{N}} y_k$$

If I want to retrieve $|y_k\rangle$ I have to repeat the algorithm $\frac{1}{\epsilon^2}$ times to approximate

$|y_k\rangle$ with precision ϵ

Phase estimation

$$U|\varphi\rangle = e^{i\theta} |\varphi\rangle$$

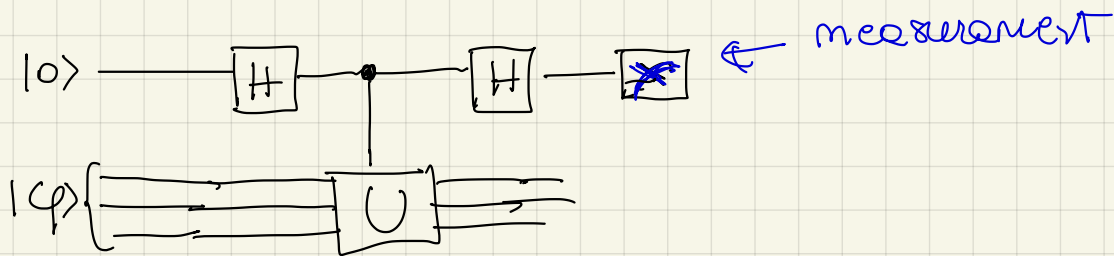
U has an eigenvector equal to $|\varphi\rangle$

We would like to remove the phase ϑ

— you have U

— you know $|\varphi\rangle$ $|\varphi\rangle \equiv \boxed{U} \equiv e^{i\vartheta} |\varphi\rangle$

the Hadamard test



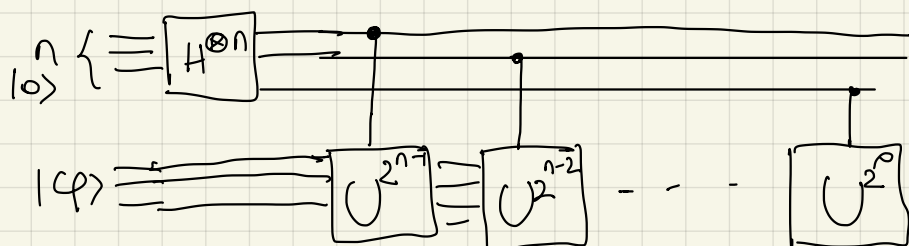
$$|0\rangle|\varphi\rangle \xrightarrow{H \otimes I} \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |\varphi\rangle \xrightarrow{C-U} \frac{|0\rangle|\varphi\rangle}{\sqrt{2}} + \frac{e^{i\vartheta}}{\sqrt{2}} |1\rangle|\varphi\rangle$$

$$\xrightarrow{H \otimes I} \left(\frac{1+e^{i\vartheta}}{\sqrt{2}} \right) |0\rangle|\varphi\rangle + \left(\frac{1-e^{i\vartheta}}{\sqrt{2}} \right) |1\rangle|\varphi\rangle$$

$$\vartheta = 2^\circ \rightarrow \left| \frac{1+e^{i\vartheta}}{\sqrt{2}} \right|^2 = \text{pr}(\text{measurin } 0) = 0.999\dots 61$$

$$\left| \frac{1-e^{i\vartheta}}{\sqrt{2}} \right|^2 = \text{pr}(\text{measurin } 1) = 2.77 \cdot 10^{-3}$$

We can use n qubits to compute encode the binary representation of ϑ



$$U^{2^s} |\varphi\rangle = e^{i\theta 2^s} |\varphi\rangle$$

$$|0\rangle^{\otimes n} |\varphi\rangle \xrightarrow{H^{\otimes n}} \frac{1}{\sqrt{N}} (|0\rangle + |1\rangle)^{\otimes n} |\varphi\rangle \xrightarrow{C_1 - U^{2^{n-1}}}$$

$$\frac{1}{\sqrt{N}} (|0\rangle + e^{i\theta 2^{n-1}} |1\rangle) (|0\rangle + |1\rangle)^{\otimes (n-1)} |\varphi\rangle$$

$$\vdots$$

$$\frac{1}{\sqrt{N}} (|0\rangle + e^{i\theta 2^{n-1}} |1\rangle) \otimes (|0\rangle + e^{i\theta 2^{n-2}} |1\rangle) \dots \otimes (|0\rangle + e^{i\theta 2^0} |1\rangle) |\varphi\rangle$$

$$\theta = (\theta_1 \theta_2 \dots \theta_n) \quad \theta = \theta_1 2^{-1} + \theta_2 2^{-2} \dots + \theta_n 2^{-n}$$

$$2^{n-1} \theta = \theta_1 2^{n-2} + \theta_2 2^{n-3} \dots + \theta_n 2^{-1}$$

$$e^{2\pi i \theta 2^{n-1}} = e^{2\pi i (\theta_n)}$$

QPE



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